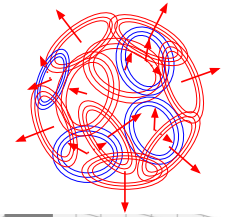
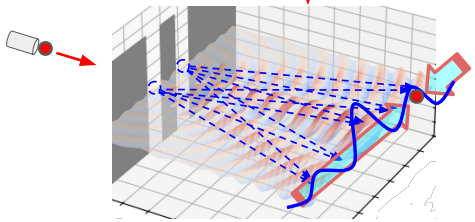
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -





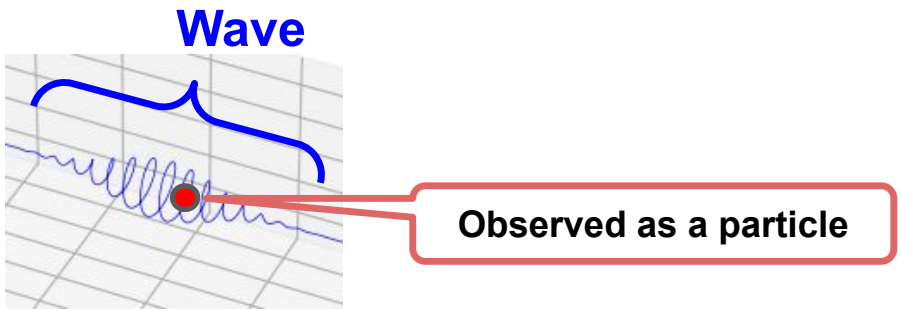
Chap.4:
The Mechanism of Wave Function Collapse

1. Quantum Duality

Quanta, such as photons and electrons, possess both particle and wave properties, a phenomenon known as quantum duality.

Before being observed, a quantum behaves as a wave spreading through space; however, when it is observed, it is detected as a single particle.

In quantum mechanics, this wave spreading through space is represented by a wave function.



1. Quantum Duality

The process in which the wave function "breaks" upon observation, causing a quantum that was a wave to be detected as a particle, is called wave function collapse.

There are various interpretations regarding this collapse, and the issue remains unresolved in modern physics.

In R-space Theory, a quantum is defined as a combination of rotation vectors, which represent angular momentum.

This theory explains the nature of the wave function and its collapse through these rotation vectors and their projections.

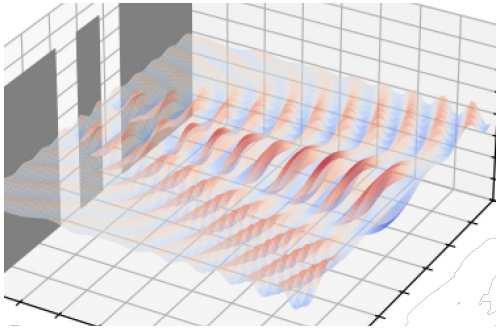
2. The Mysterious Interference of the Quantum Double-Slit Experimen

Wave interference is a common natural phenomenon seen in light, sound, and water waves.

A wave is a superposition of smaller waves.

When waves pass through a double slit, the path length to different positions varies, causing the phases of the waves to shift.

This results in a pattern of varying wave intensity depending on the position.

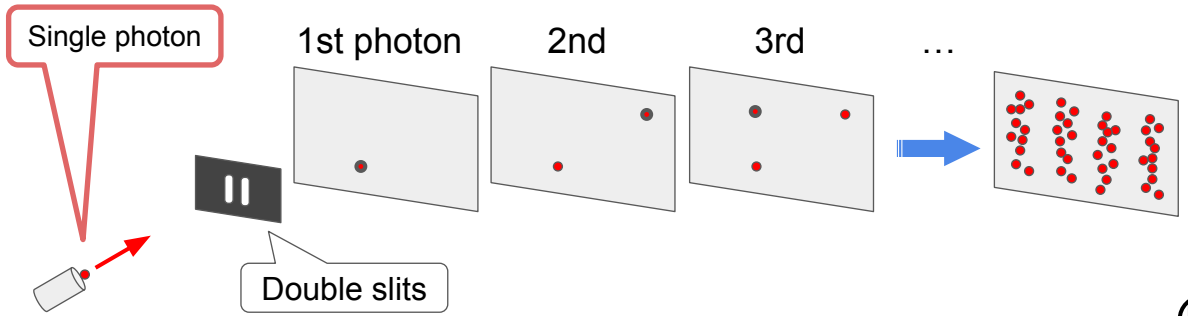


2. The Mysterious Interference of the Quantum Double-Slit Experimen

Quantum waves behave similarly; when quanta pass through a double slit, an interference fringe of quantum density forms on the observation screen.

The mysterious aspect of quantum waves, however, is that interference fringes appear even when quanta are sent through the double slit one at a time.

It is as if a single quantum passes through both slits simultaneously and interferes with itself.

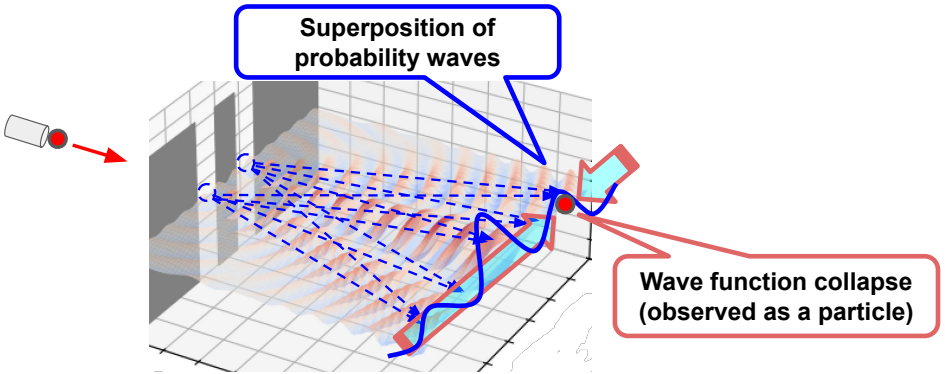


2. The Mysterious Interference of the Quantum Double-Slit Experimen

Quantum mechanics explains this behavior as the superposition of probability waves represented by the wave function.

Probability waves pass through both slits, creating an interference pattern on the screen.

Then, the wave function collapses, and the quantum is observed at a specific location.



2. The Mysterious Interference of the Quantum Double-Slit Experimen

The probability of observation is higher where the amplitude of the probability wave is larger.

Yet, as mentioned previously, interpretations differ on why the wave function collapses and results in a particle-like observation.

In R-space Theory, there is only rotation and phase.

Spacetime is generated from this rotation and phase, giving rise to quantum duality and the mysterious phenomena seen in experiments like the double-slit experiment.

3. Spacetime Created by Rotation and Phase

The world consists of combinations of rotation vectors representing rotation and phase in R-space.

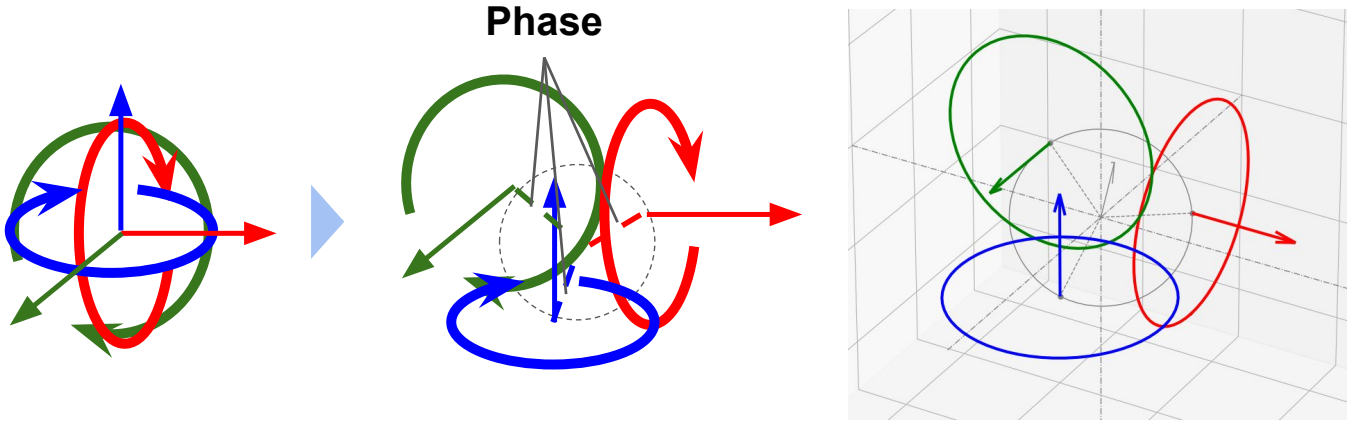
However, because our bodies are also composed of rotation vectors, we cannot directly perceive that rotation and phase.

Instead, we recognize the world as spacetime (time + three-dimensional space), which is a projection of R-space.

3. Spacetime Created by Rotation and Phase

In R-space, a photon is a combination of rotation vectors with components in three orthogonal directions (x, y, and z).

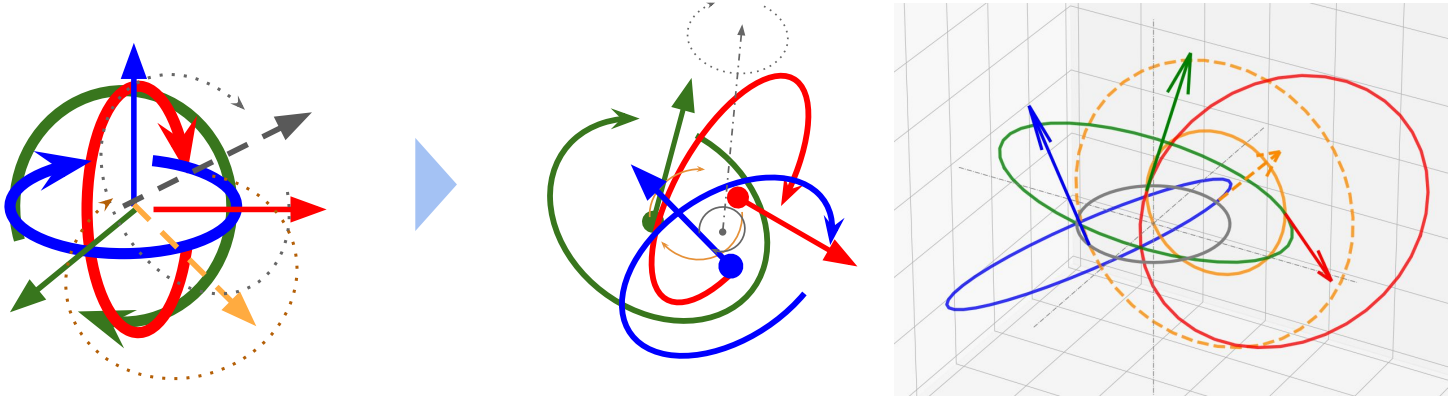
Vector x acts as the phase for vectors y and z; similarly, y acts as the phase for z and x, and z acts as the phase for x and y.



3. Spacetime Created by Rotation and Phase

An electron is also a combination of three orthogonal rotation vectors.

However, as explained in Chapter 2, because its rotational energy is high, the three rotation vectors form pairs that undergo revolution and precession.

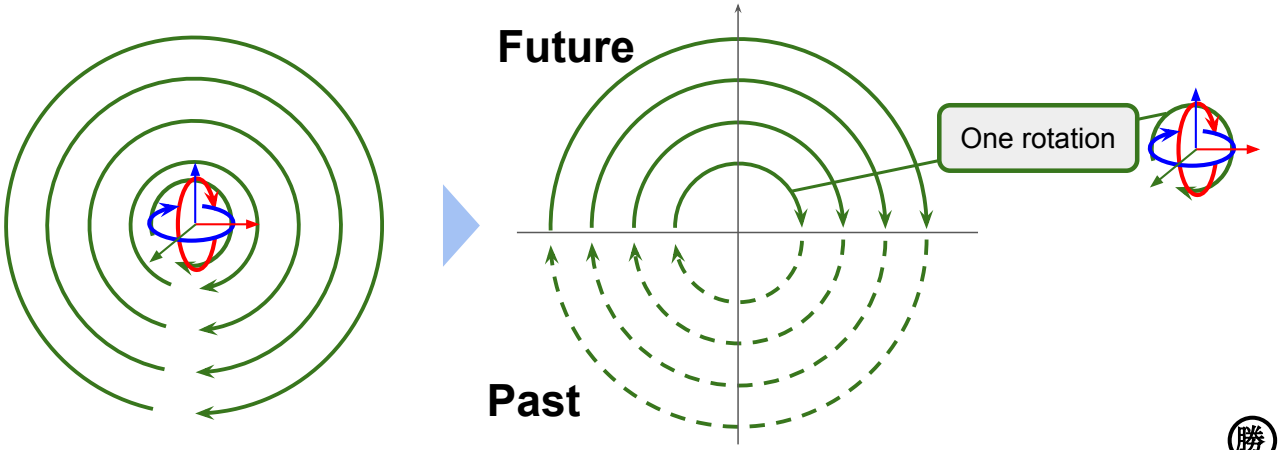


3. Spacetime Created by Rotation and Phase

To visualize the number of phase rotations, concentric circles are used where the radius increases with each rotation.

To correspond with our perceived spacetime, these are divided into a "future" (upper) and "past" (lower) half.

One semi-circle corresponds to one full rotation of the rotation vector's phase.

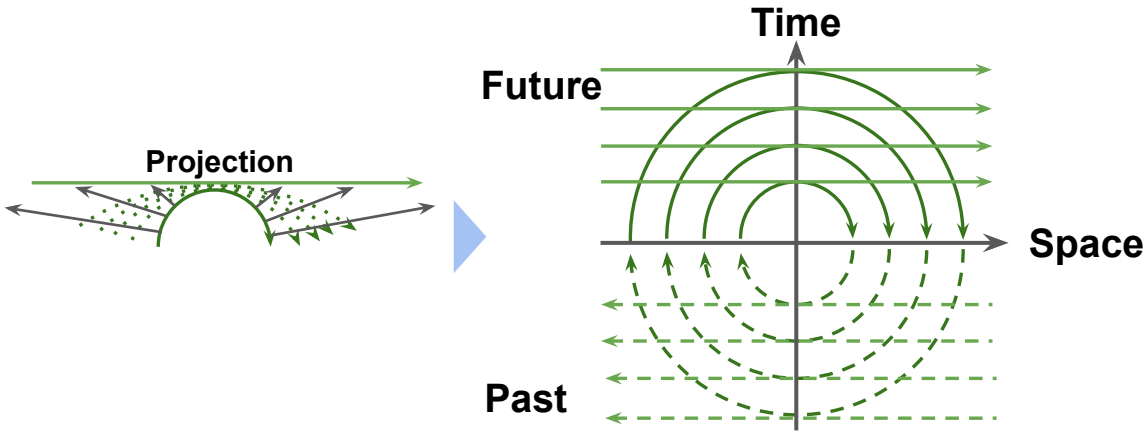


3. Spacetime Created by Rotation and Phase

We perceive these semi-circles as being projected into straight space.

This is the state where R-space is projected into spacetime.

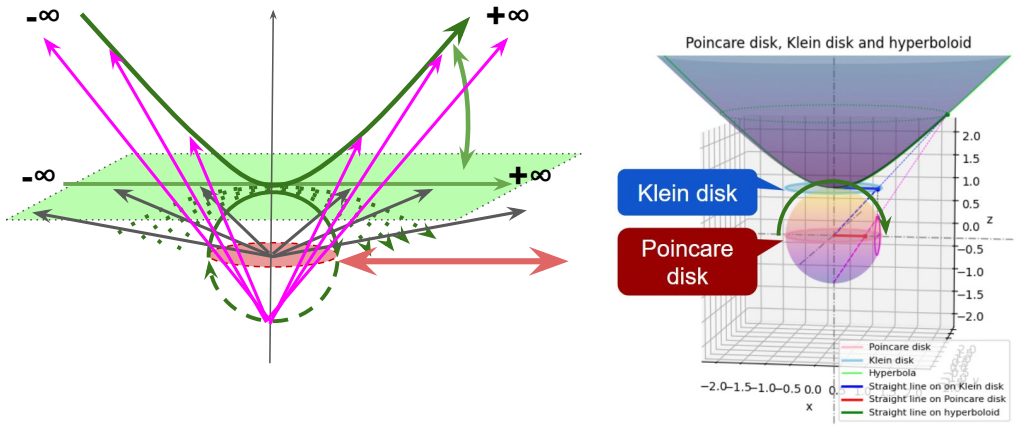
It is analogous to how humans living on a spherical Earth perceive their environment as a flat plane.



3. Spacetime Created by Rotation and Phase

This projection is mathematically similar to a hemisphere (or semi-circle) being projected via a Poincaré disk onto an infinitely spreading hyperboloid (or hyperbola).

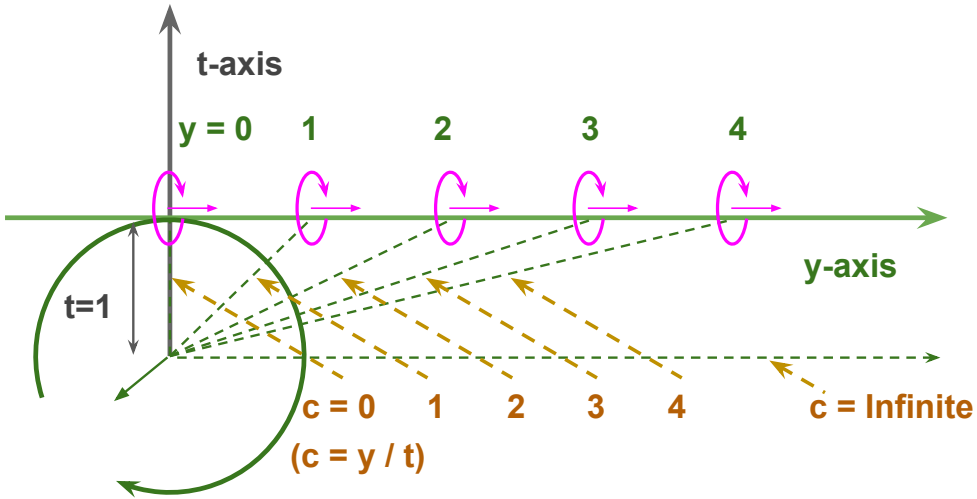
In the case of R-space's projection into space, it becomes an infinite plane (or straight line).



3. Spacetime Created by Rotation and Phase

Furthermore, the slope in the projected spacetime corresponds to velocity, much like in Minkowski space.

Consequently, a quantum (rotation vector) is observed to progress through spatial coordinates according to the slope of its phase—that is, its velocity.



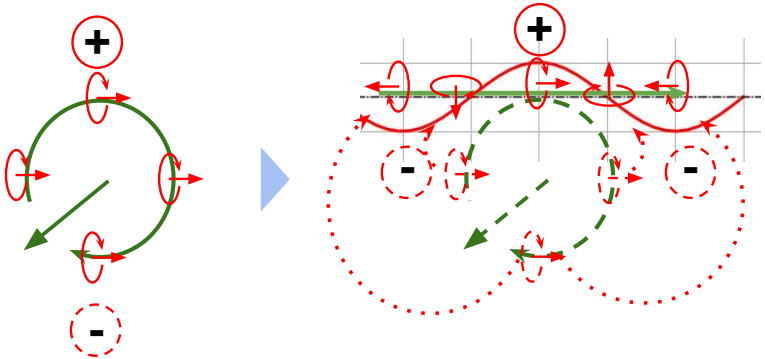
3. Spacetime Created by Rotation and Phase

When the photon's phase semi-circle is "unrolled" to project R-space onto the spatial axis, the orientation of the rotation vector alternates (e.g., right, up, left, down).

If right is positive and left is negative, it repeats a cycle of plus and minus.

This corresponds to the photon's wave nature.

Thus, wave properties are generated from the rotation vector and its phase.

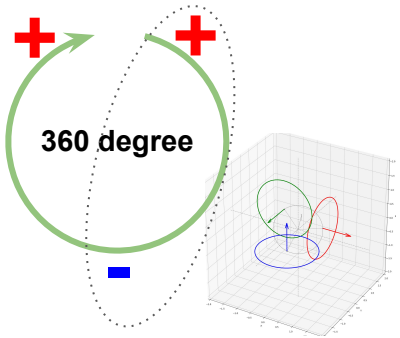


3. Spacetime Created by Rotation and Phase

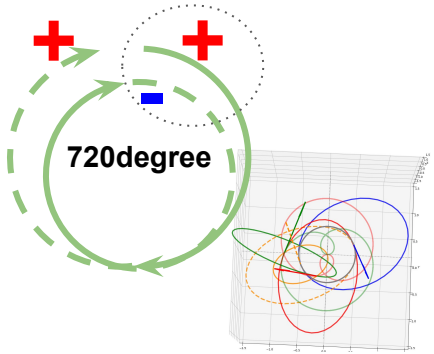
Photons have spin 1 (360° rotational symmetry), while electrons have spin 1/2 (720° rotational symmetry).

A photon completes one rotation in 360°, whereas an electron requires 720°.

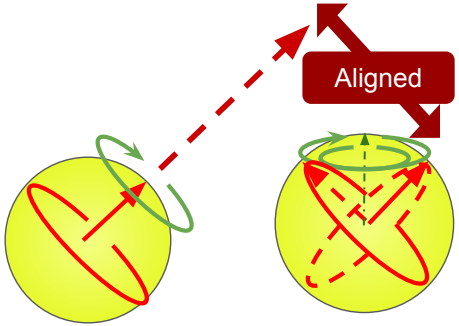
For an electron's rotation vector to absorb the energy of a photon's rotation vector, their axes of rotation must align.



Photon



Electron

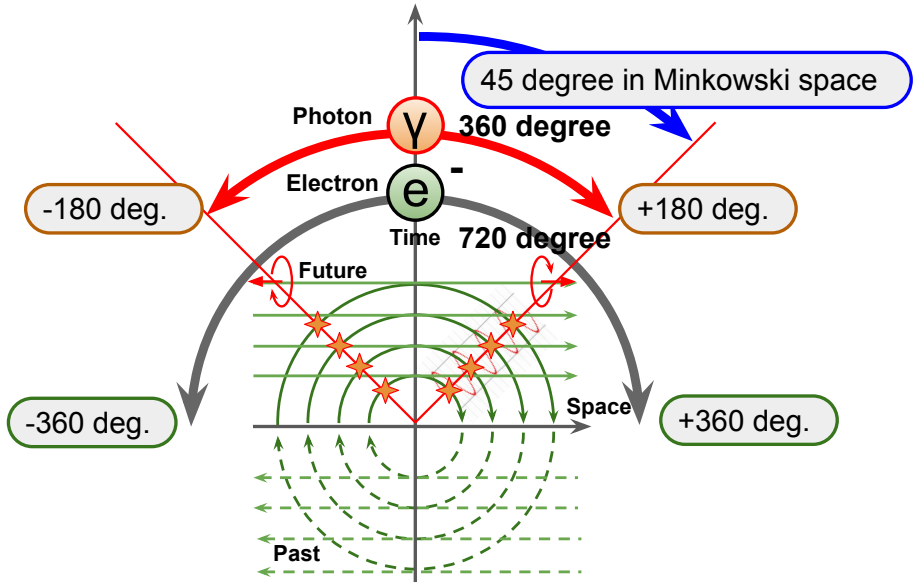


3. Spacetime Created by Rotation and Phase

From the electron's perspective, the R-circle's semi-circle 180°(±90°) corresponds to a range of 720°(±360°) .

Therefore, the position at half the semi-circle 90°(±45°) corresponds 360° (±180°).

As a result, photons and electrons always interact at a phase of 45° from the time axis.

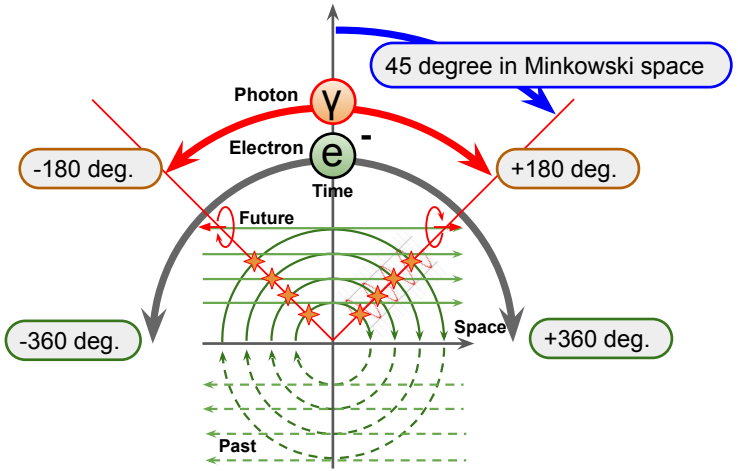


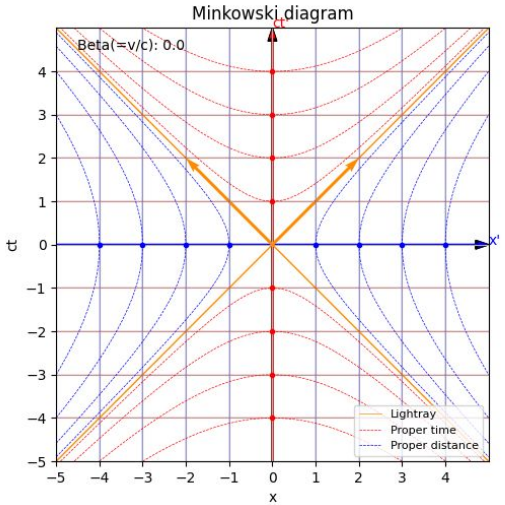
3. Spacetime Created by Rotation and Phase

A phase (slope) of 45° in Minkowski space corresponds to the speed of light.

This is why we perceive R-space as Minkowski space.

In this manner, the spacetime we recognize is generated from the rotation and phase of R-space.



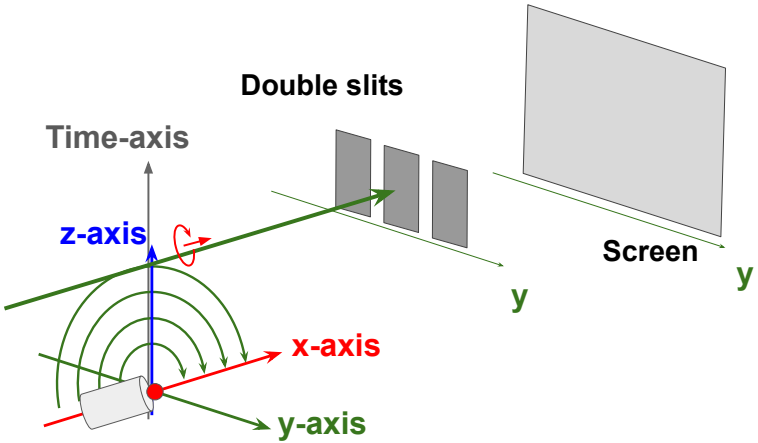


4. Double-Slit Experiment and Wave Function Collapse

The double-slit experiment requires an additional dimension.

Suppose two slits are aligned along the y-axis, which is orthogonal to the x-axis.

Since we cannot visualize all four dimensions—one time dimension and three spatial dimensions—the z-axis is drawn to represent the time axis.

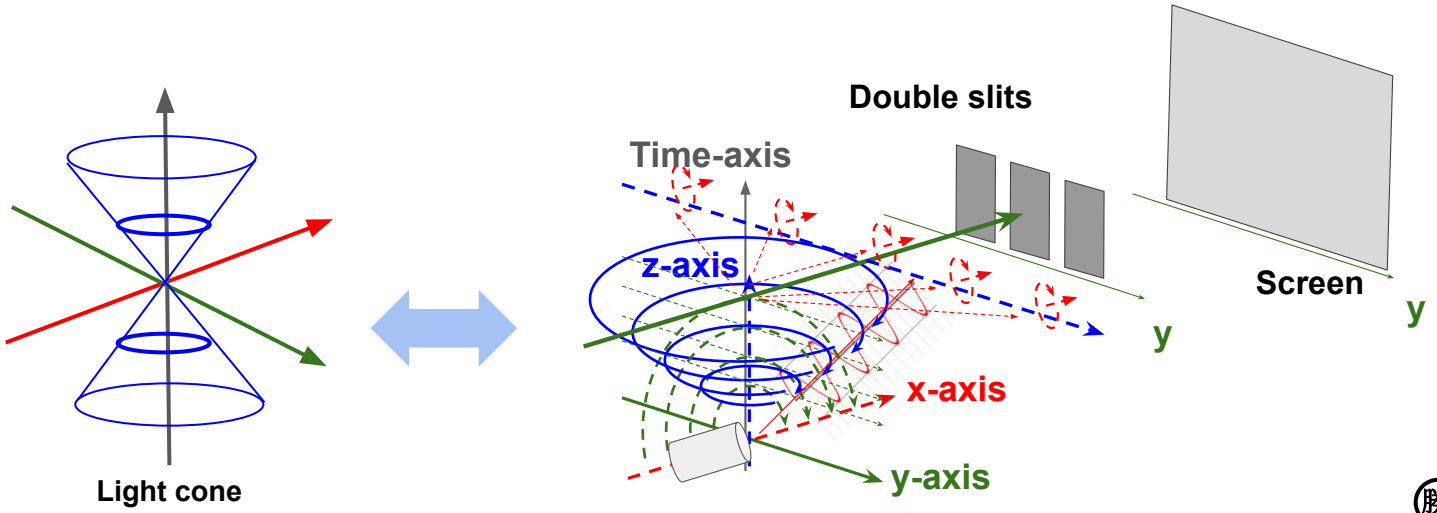


4. Double-Slit Experiment and Wave Function Collapse

When a photon is emitted from the origin, rotation vector z acts as the phase for projections along the y-axis (along the slits and screen).

As the phase rotation of vector z progresses, its phase circle expands.

This corresponds to the light cone in Minkowski space.



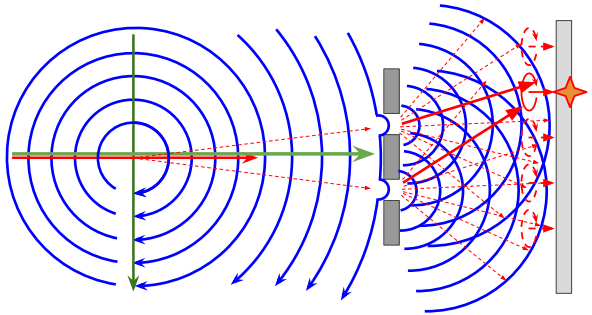
4. Double-Slit Experiment and Wave Function Collapse

Viewed from the z-axis direction, these appear as concentric circles.

As the phase rotation continues, the phase circle reaches the double slits, and new phase circles begin to spread from each slit.

These expanding phase circles from the two slits overlap and interfere with each other.

This corresponds to the superposition of wave functions in quantum mechanics.

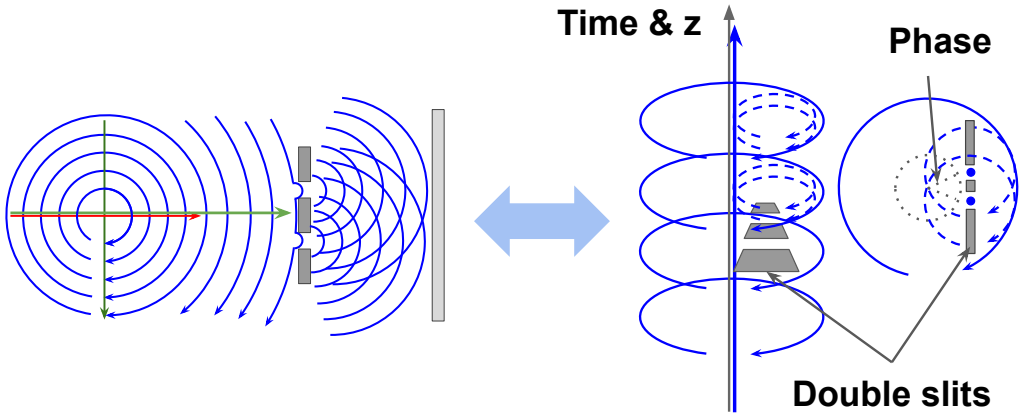


4. Double-Slit Experiment and Wave Function Collapse

While diagrams show the two phase circles as separate, there is no distance in R-space as we perceive it; there is only phase and rotation.

Separate phase circles in the diagram mean the phases are split in R-space.

However, these two phase circles are actually overlapping as a single vortex in R-space.



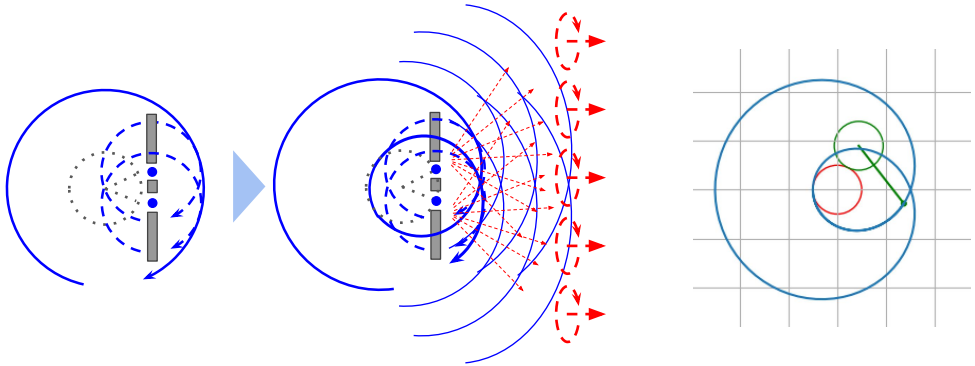
4. Double-Slit Experiment and Wave Function Collapse

The phase of rotation vector x is projected onto the observation screen according to the phase of these overlapping vortices.

The probability of it being projected by either vortex is fifty-fifty.

At this moment, the way the rotation vector x moves around the two phase circles looks as shown in the following figure.

It is like a leaf drifting over multiple vortices on the surface of water.

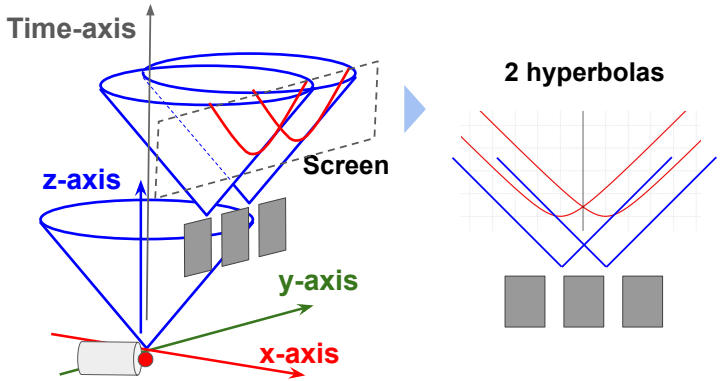


4. Double-Slit Experiment and Wave Function Collapse

To explain the interference, one must consider the wave nature of the rotation vector relative to the time axis.

Viewed via light cones, the cone splits into two at the position of the double slits.

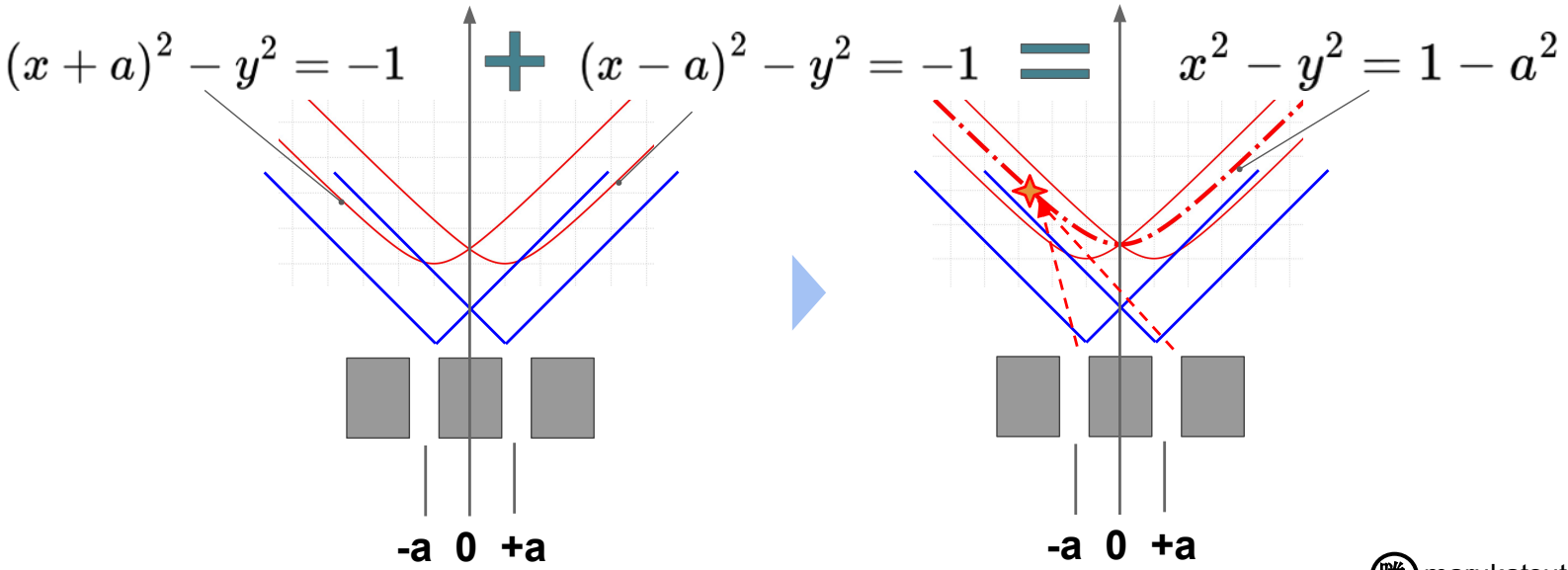
The photon reaches the screen at the intersection of the screen and these two light cones, which forms hyperbolas.



4. Double-Slit Experiment and Wave Function Collapse

If the distance from the center of each slit is a, the equations for the two hyperbolas are ”(x - a)^2 - y^2 = -1” and ”(x + a)^2 - y^2 = -1”.

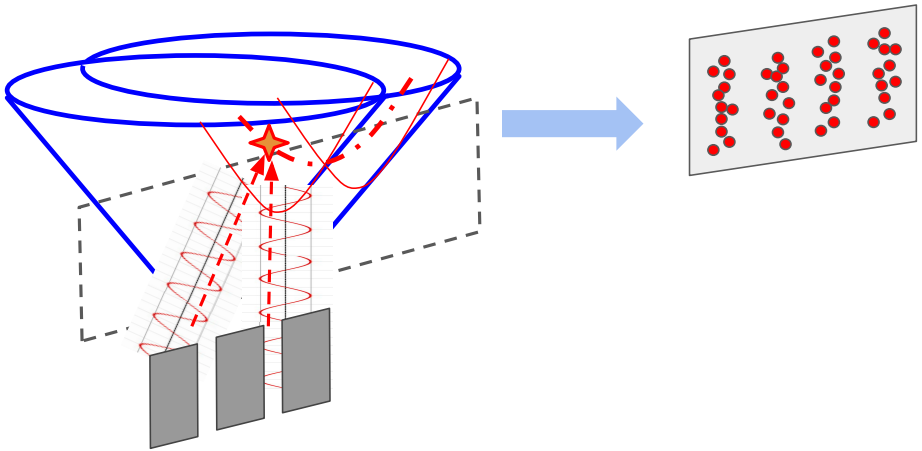
Adding these gives ”x^2 - y^2 = 1 - a^2”, which is also a hyperbola.



4. Double-Slit Experiment and Wave Function Collapse

The photon reaches the screen at some point on this combined hyperbola depending on the phase of the vector z vortex.

At that moment, the difference in path lengths between the two hyperbolas causes a phase difference, resulting in interference fringes.



4. Double-Slit Experiment and Wave Function Collapse

The photon is interfering with the wave generated by its own rotation vector.

This is similar to the concept of pilot waves in De Broglie–Bohm theory.

However, in this theory, what "rides" the wave is not a particle, but a rotation vector representing angular momentum.

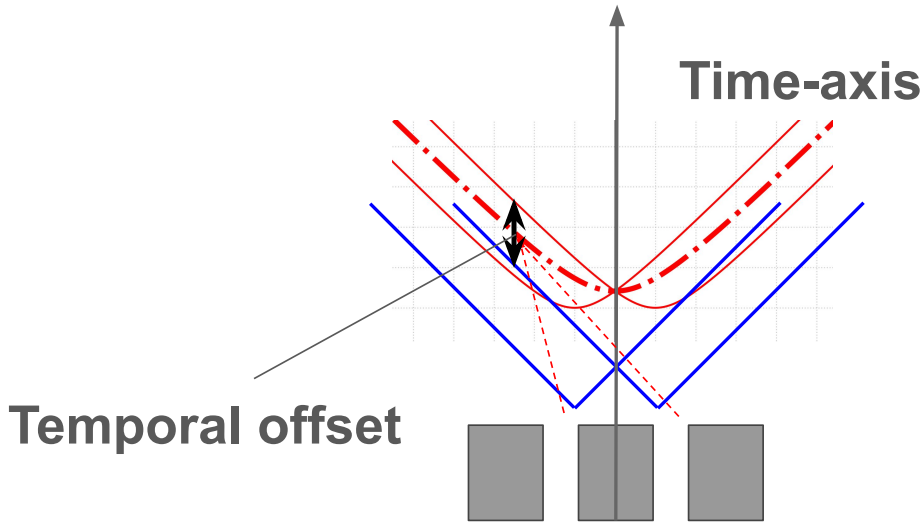
Furthermore, the pilot wave does not actually spread through space; rather, the phase of the rotation vector is simply rotating.

Because we can only perceive this as a projection into space, it is observed as a spreading wave.

5. Overlapping Spacetime

There is a curious point regarding the addition of the two hyperbolas: except at the center, the time the photon reaches the screen differs between the two hyperbolas (two phase circles).

This is because time does not "flow" in the way we feel it does.

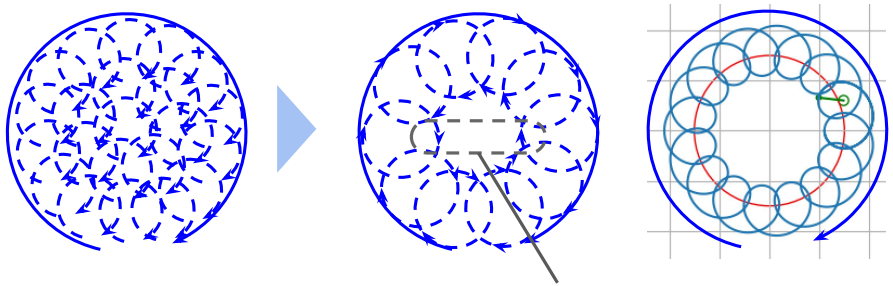


5. Overlapping Spacetime

In the double-slit experiment, the vortex (phase) of the rotation vector splits into two.

However, even the original vortex before splitting is a collection (overlap) of smaller vortices.

Because the directions of rotation cancel each other out inside the phase circle, the whole assembly behaves as one large vortex.

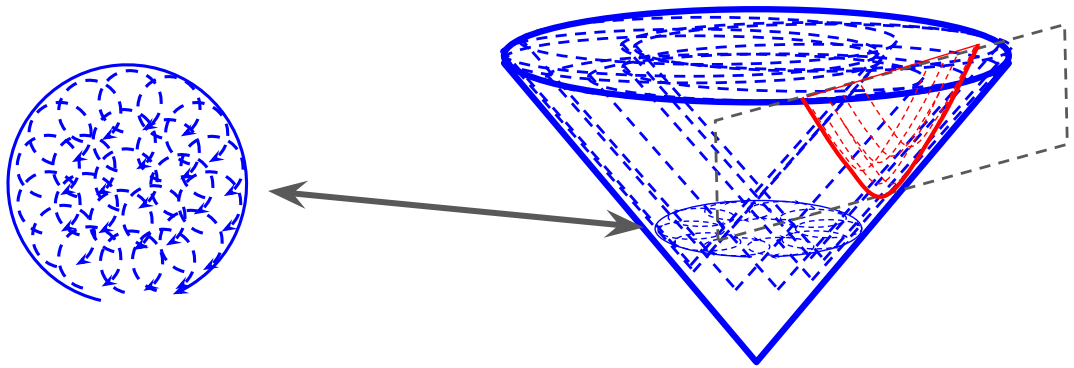


Cancel out

5. Overlapping Spacetime

Since the phase circle is a stack of small vortices, the light cone spreading into spacetime is also a stack of small light cones.

Consequently, the hyperbola—the projection of that light cone—is a collection of small hyperbolas.

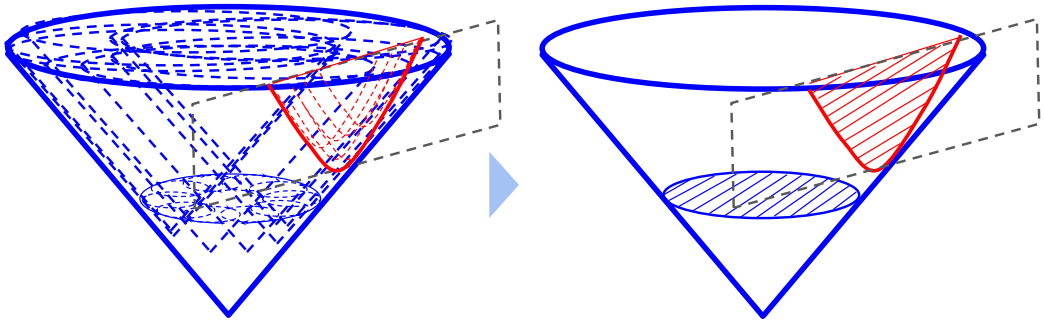


5. Overlapping Spacetime

The phase circle of a photon's rotation vector is not just a circumference, but a surface including its interior.

Likewise, its projection, the hyperbola, is a surface rather than a line.

However, because the interior regions cancel out, only the edges are observed as a photon.



5. Overlapping Spacetime

Time is not flowing; it is stacked. Space, too, does not simply "exist"; it is stacked.

What we recognize (observe) as a photon occurs at the edge of this stacked time and space.

Therefore, phase differences also stack and are observed as interference.

The photon is detected at some point along the edge of this stacked spacetime.

Because time and space are stacked, the mysterious phenomenon occurs where a photon passes through two slits simultaneously without splitting and is observed at a single point like a particle.

This is the mechanism of wave function collapse.

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This is all for my presentation.

Thanks for watching!

